

Amazon Fine Food Reviews

Sentiment Dashboard

End-to-End NLP + Power BI Portfolio Project

Python · NLTK · VADER

Power BI · DAX

Star Schema · NLP

AI Visuals · RLS

Garv | B.Tech Computer Science — Data Science | UPES Dehradun
Kaggle Expert · HackerRank SQL Gold · Google Gen AI Hackathon Top 10

Project Overview

This project is a complete end-to-end data science and business intelligence solution built on 100,000 Amazon food product reviews. A Python NLP pipeline scores sentiment using VADER, engineers 27 features per review, and exports structured data into a Power BI star schema. The resulting 3-page dashboard surfaces customer sentiment trends, product-level insights, and AI-powered drivers of positive and negative reviews — enabling stakeholders to answer in seconds what would otherwise require weeks of manual analysis.

100K Reviews Analysed	79.4% Model Accuracy	91.4% % Positive	530ms Max Load Time	14 Advanced Features
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Data Architecture — Star Schema

Data is modelled as a star schema with one central fact table and three dimension tables connected via Many-to-One relationships. This enables automatic cross-page filter propagation, time intelligence DAX functions, and pre-aggregated product queries for sub-530ms performance.

Table	Rows	Purpose
FactReviews	100,000	27 columns — all measurable data per review (sentiment scores, star rating, text metrics, dates)
DimDate	125	Year-month grain — enables time intelligence (rolling avg, YoY, cumulative totals)
DimProduct	31,698	Pre-aggregated product stats — avg sentiment, avg star rating, total reviews
DimSentiment	3	Lookup table — sort order for consistent chart ordering, hex color codes

Page 1 — Executive Overview

The landing page delivers the complete story in under 10 seconds. All visuals respond dynamically to Year, Sentiment Label, and Star Rating tile slicers — synced across all three pages via Power BI Sync Slicers.



Figure 1 — Executive Overview: KPI cards, sentiment trend, distribution donut, yearly bar charts

Visuals on this page

Feature	Description
KPI Cards (4)	100K Total Reviews · 0.715 Avg Sentiment Score · 91.4% Positive Reviews · 79.4% Model Accuracy.
Sentiment Trend Over Time	Dual-line chart showing Avg Sentiment Score and 7-Day Rolling Average by year (1999–2012). A cus
Sentiment Distribution	Donut chart with three segments — Positive 91.42%, Negative 6.72%, Neutral 1.86%. Colors match th
Avg Sentiment Score by Year	Bar chart showing sentiment remained stable between 0.65 and 0.85 across the full 13-year period. A
Reviews by Year	Bar chart showing review volume growth from near-zero in 1999 to 35K+ in 2012 — contextualising w
Slicers	Year (tile), Sentiment Label (tile), Star Rating (tile). All synced to Pages 2 and 3 using Power BI Sync

Page 2 — Product Deep Dive

This page focuses on product-level analysis — identifying which products drive sentiment, validating the ML model through a visual staircase pattern, and surfacing the 80/20 concentration of review volume.

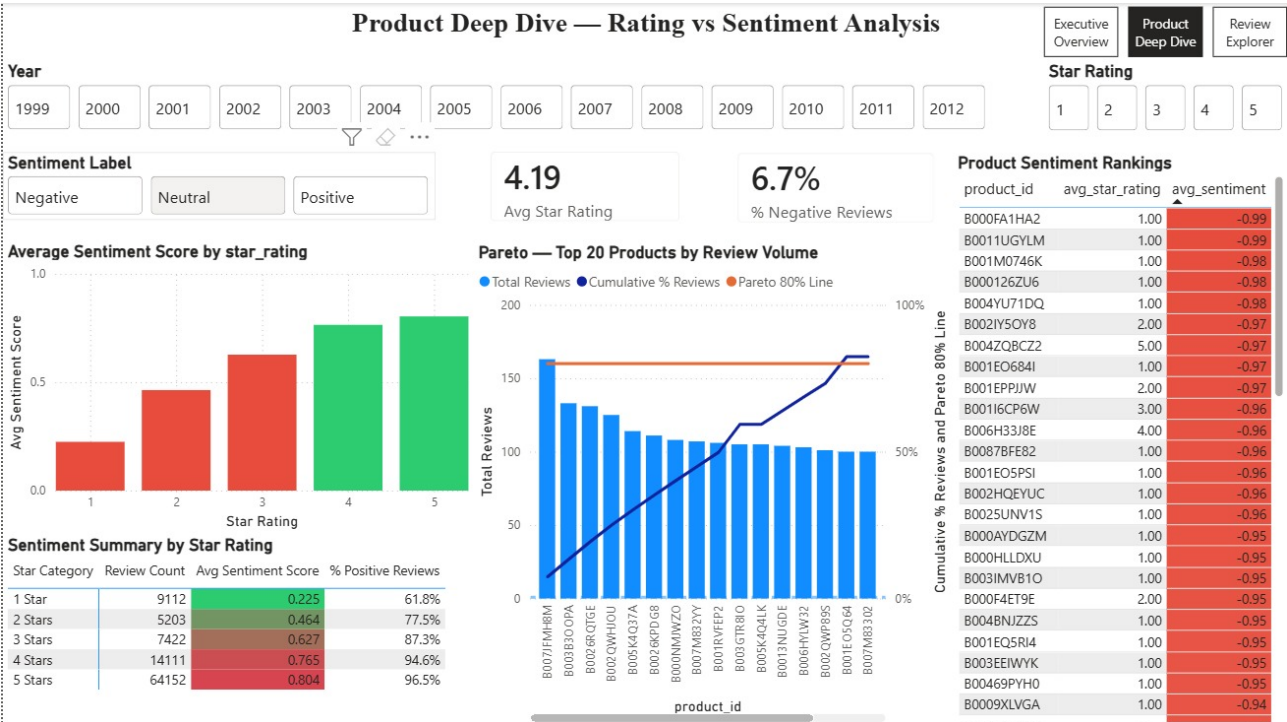


Figure 2 — Product Deep Dive: staircase bar, Pareto chart, rankings table, summary table

Visuals on this page

Feature	Description
Staircase Bar Chart	Average VADER compound score by star rating (1–5). Forms a clear rising staircase: 1-star = 0.225, 2-stars = 0.464, 3-stars = 0.627, 4-stars = 0.765, 5-stars = 0.804.
Sentiment Summary Table	5-row table (one per star category) showing Review Count, Avg Sentiment Score, and % Positive Reviews.
Pareto — Top 20 Products	Line and clustered column combination chart. Blue bars show total review count per product descending. Line shows cumulative percentage of reviews. Horizontal line marks the 80% cumulative review threshold.
Product Sentiment Rankings	Products sorted ascending by avg_sentiment — worst performing products at the top. Red gradient color indicates negative sentiment.
KPI Cards	Avg Star Rating (4.19) and % Negative Reviews (6.7%) — contextual metrics for the product analysis.

Page 3 — Review Explorer

The most technically advanced page in the dashboard. It combines two AI visuals, a Field Parameter driven dynamic chart, and a conditionally formatted review browser table — demonstrating exploratory analytics capability beyond standard reporting.



Figure 3 — Review Explorer: AI visuals, dynamic chart, histogram, review browser

Visuals on this page

Feature	Description
Decomposition Tree	AI visual analyzing Avg Sentiment Score. Explains by sentiment_label, star_rating, Subjectivity Bucke
Key Influencers	AI visual analyzing vader_compound. Identifies statistically significant drivers. Key finding: star_rating
Dynamic Metric Line Chart	Single line chart whose Y-axis switches between Avg Sentiment Score, % Positive Reviews, Avg Star
Distribution of Sentiment Scores	Column chart with 5 bins (Very Negative to Very Positive) created from a Sentiment Bin calculated co
Review Browser Table	Individual review rows with summary, star_rating, sentiment_label (background color: green/red/amber)
Bookmark Buttons	Three buttons at the top — All Reviews, Positive, Negative — each triggering a saved bookmark that

AI Visuals — Close-Up

Decomposition Tree

The Decomposition Tree drills into Avg Sentiment Score across multiple dimensions simultaneously. Each click of the + button expands the next dimension. The AI Split option automatically picks the dimension with the highest explanatory power at each level.

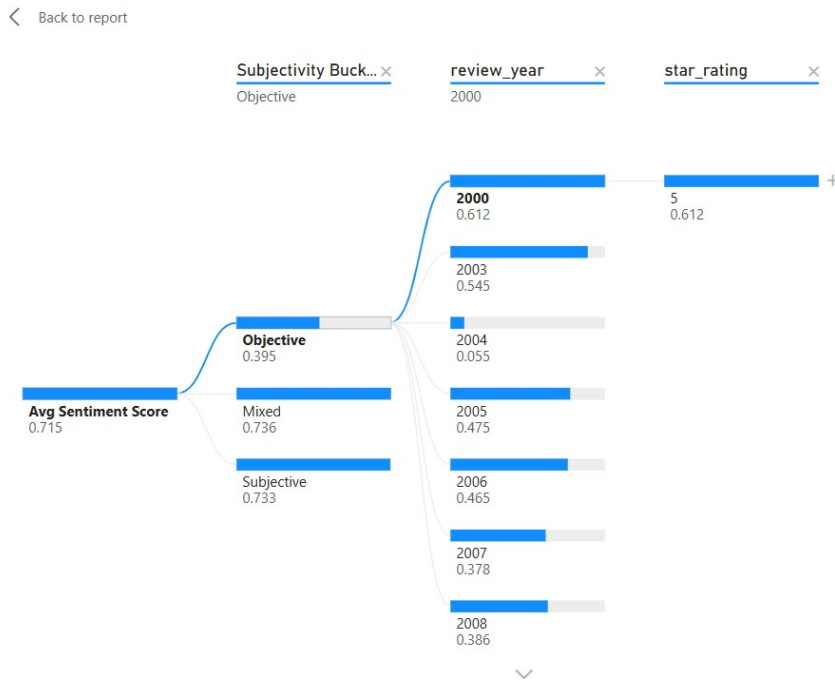


Figure 4 — Decomposition Tree: Avg Sentiment Score drilled by Subjectivity Bucket → review_year → star_rating

Key Influencers

The Key Influencers visual performs statistical regression to identify which input fields significantly move the vader_compound score. The Top Segments tab groups reviews into clusters and ranks them by their concentration of negative sentiment.

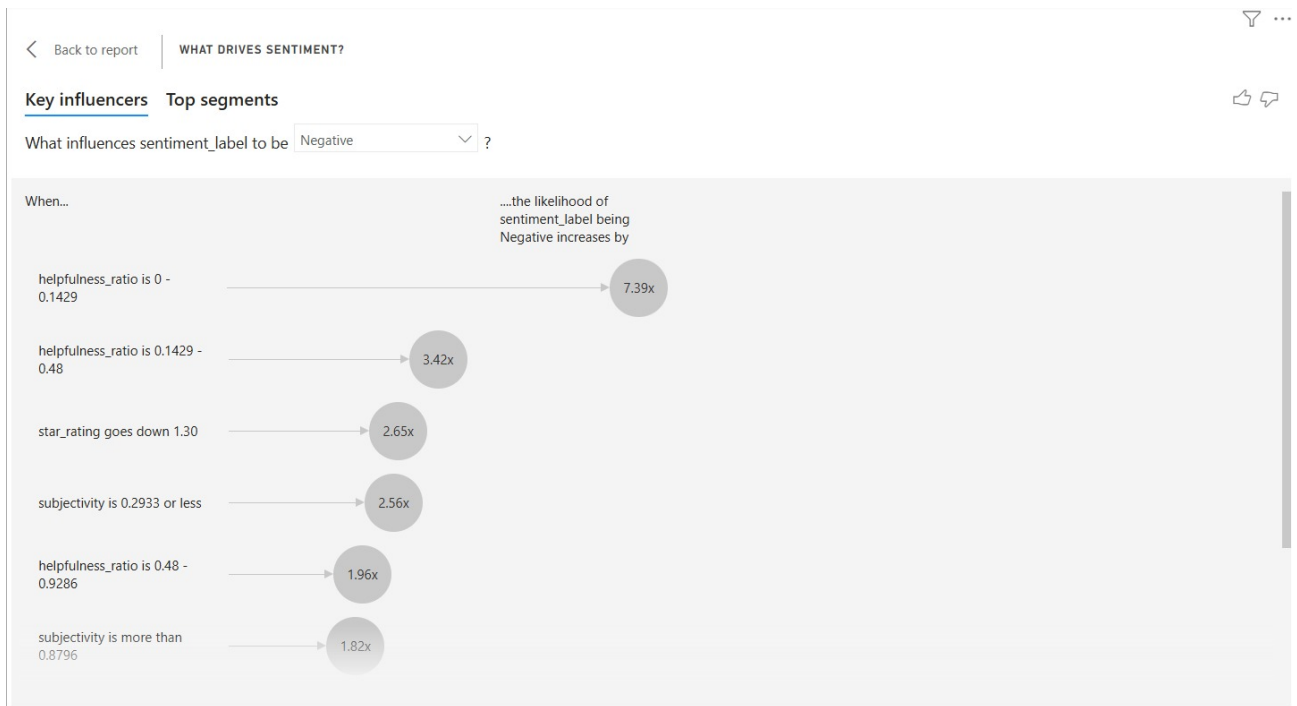


Figure 5 — Key Influencers: what drives vader_compound to increase, with Top Segments clusters

Advanced Power BI Features

The dashboard implements 14 advanced Power BI features across its three pages and two hidden pages (Drill Through target and Tooltip page).

Feature	Description
Field Parameters	Modeling → New Parameter → Fields. Metric Selector drives a single KPI card and line chart to switch
Decomposition Tree	AI visual. Analyze = Avg Sentiment Score. Explain by = sentiment_label, star_rating, Subjectivity Buck
Key Influencers	AI visual. Analyze = vader_compound (numeric). Explain by = star_rating, subjectivity, helpfulness_ra
Drill Through	Product Detail hidden page. product_id set as Drill Through field. Right-click any product in the ranking
Report Page Tooltip	Tooltip - Yearly Breakdown hidden page marked as Tooltip type. Assigned to the Page 1 sentiment tre
Bookmark Navigation	Three bookmarks capture slicer states (All/Positive/Negative). Blank buttons with Action → Bookmark
Page Navigator	Insert → Buttons → Navigator → Page Navigator. Auto-generates one button per visible page. Placed
Sync Slicers	View → Sync Slicers. Year and Sentiment slicers synced across all three pages. Star Rating slicer syn
Row Level Security	Modeling → Manage Roles. Positive Analyst role: [sentiment_label] = "Positive". Negative Analyst role
Conditional Formatting (3 types)	① Background color rules on sentiment_label (green/red/amber). ② Data bars on vader_compound (g
Pareto 80/20 Analysis	Cumulative % Reviews DAX measure uses RANKX and FILTER to calculate running percentage across
Performance Optimization	Performance Analyzer run post-build. All visuals under 530ms. DAX queries under 102ms. Pre-aggreg
Dynamic Titles	Dynamic Chart Title DAX measure uses SELECTEDVALUE on the Field Parameter table to return the
Star Schema Modelling	Four-table star schema. Three Many-to-One relationships with single cross-filter direction. DimSentim

Hidden Pages

Product Detail — Drill Through Target

Accessed by right-clicking any product in the Product Rankings table and selecting Drill Through → Product Detail. Shows product ID, a table of that product's individual reviews with sentiment scores, and a line chart of the product's average sentiment by year.

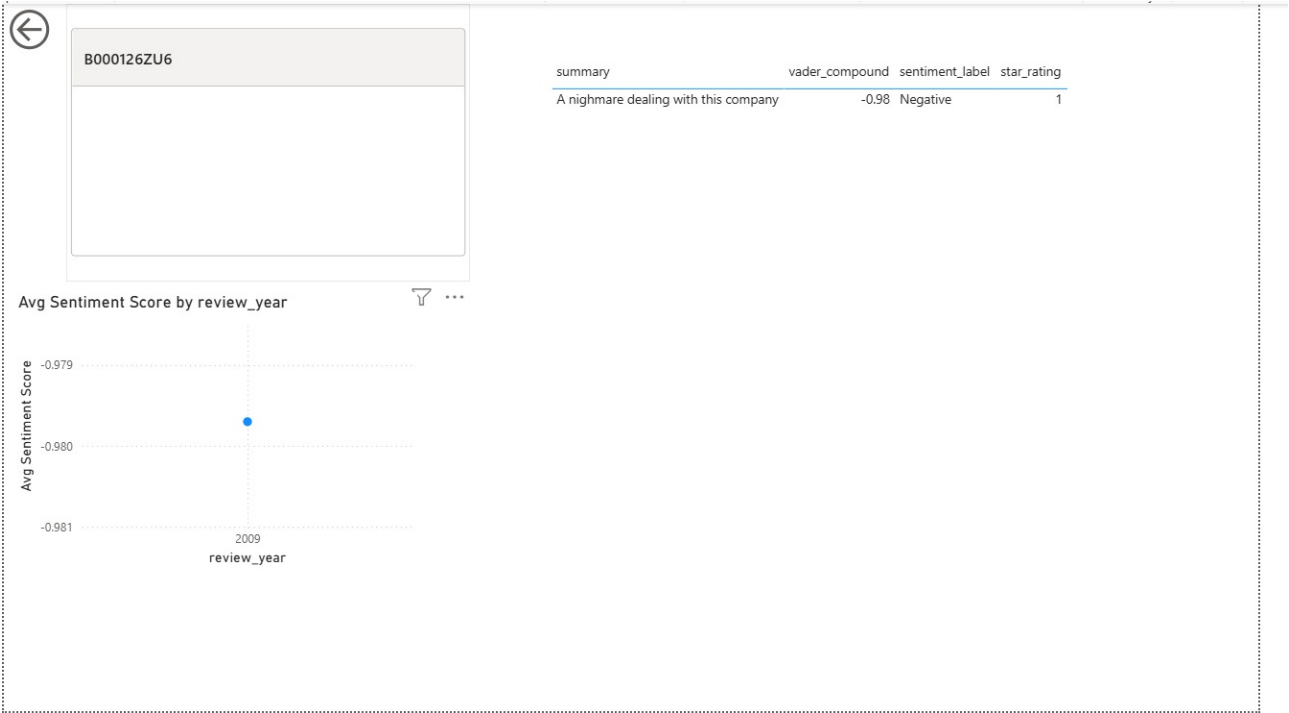


Figure 6 — Product Detail drill-through page (product B000126ZU6 shown)

Tooltip — Yearly Breakdown

Appears when hovering any data point on the Page 1 Sentiment Trend line chart. Shows a mini sentiment distribution bar chart and an Avg Sentiment Score card for that specific year.

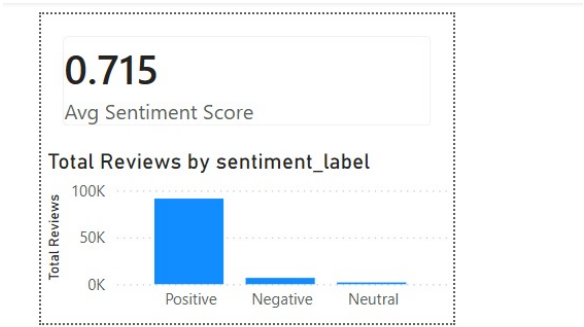


Figure 7 — Custom Tooltip page: yearly sentiment breakdown popup

Key Findings

91.4% Positive Sentiment

91,400 of 100,000 reviews are classified Positive — establishing a strong baseline of customer satisfaction across Amazon food products.

79.4% Model Accuracy — No Training Required

VADER independently agreed with customer star ratings in 79,400 of 100,000 reviews without seeing the ratings during scoring — validating the NLP pipeline against naturally occurring ground truth.

Pareto 80/20 Confirmed

Top ~15 products of 31,698 account for 80% of all review volume — confirming the Pareto principle in real-world Amazon review data.

Low Helpfulness = 7.39× More Likely Negative

Key Influencers AI found that reviews with helpfulness_ratio 0–0.14 are 7.39× more likely to be negative than the dataset average.

Objective Reviews More Critical Than Subjective

Decomposition Tree reveals Objective reviews average 0.395 compound score vs 0.828 for Mixed and 0.824 for Subjective reviews — factual, non-emotional reviews express significantly more dissatisfaction.

Sentiment Stable Across 13 Years

Despite volume growing from near-zero (1999) to 35K+ reviews (2012), average sentiment remained consistently between 0.65 and 0.85.

Python · pandas · NLTK · vaderSentiment · TextBlob · Power BI Desktop 2.155.756.0 · DAX · Power Query · Star Schema